

TWITTER SENTIMENT ANALYSIS USING MACHINE LEARNING

Classifying Twitter Sentiment for Apple and Google Products



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Data Science



Machine Learning & NLP Project

BUSINESS UNDERSTANDING

The Problem

-  Millions of customers express opinions online every day.
- Companies need to:
- Monitor brand reputation
- Detect customer dissatisfaction
- Understand public sentiment
- Improve decision-making

Project Goal

- Develop a machine learning model capable of automatically classifying tweets into:
- 😊 Positive
- 😐 Neutral
- 😞 Negative

DATASET OVERVIEW

Dataset Summary

Metric	Value
Source	Twitter
Brands	Apple & Google
Target	Sentiment
Classes	Positive, Neutral, Negative

Features

-  Tweet Text
-  Target Brand
-  Sentiment Label

DATA CLEANING & PREPARATION

Raw Tweets



Lowercase



Remove URLs



Remove Mentions



Remove Emojis



Remove Stopwords



Lemmatization



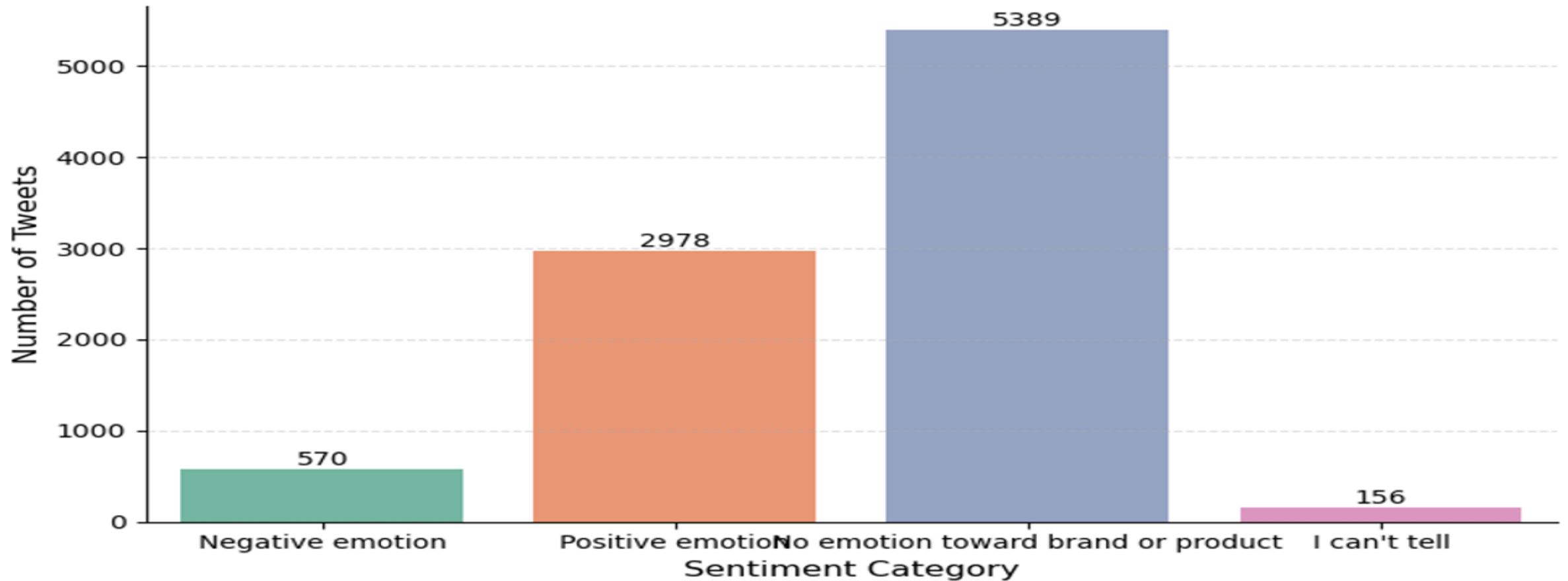
Clean Tweets

Why?

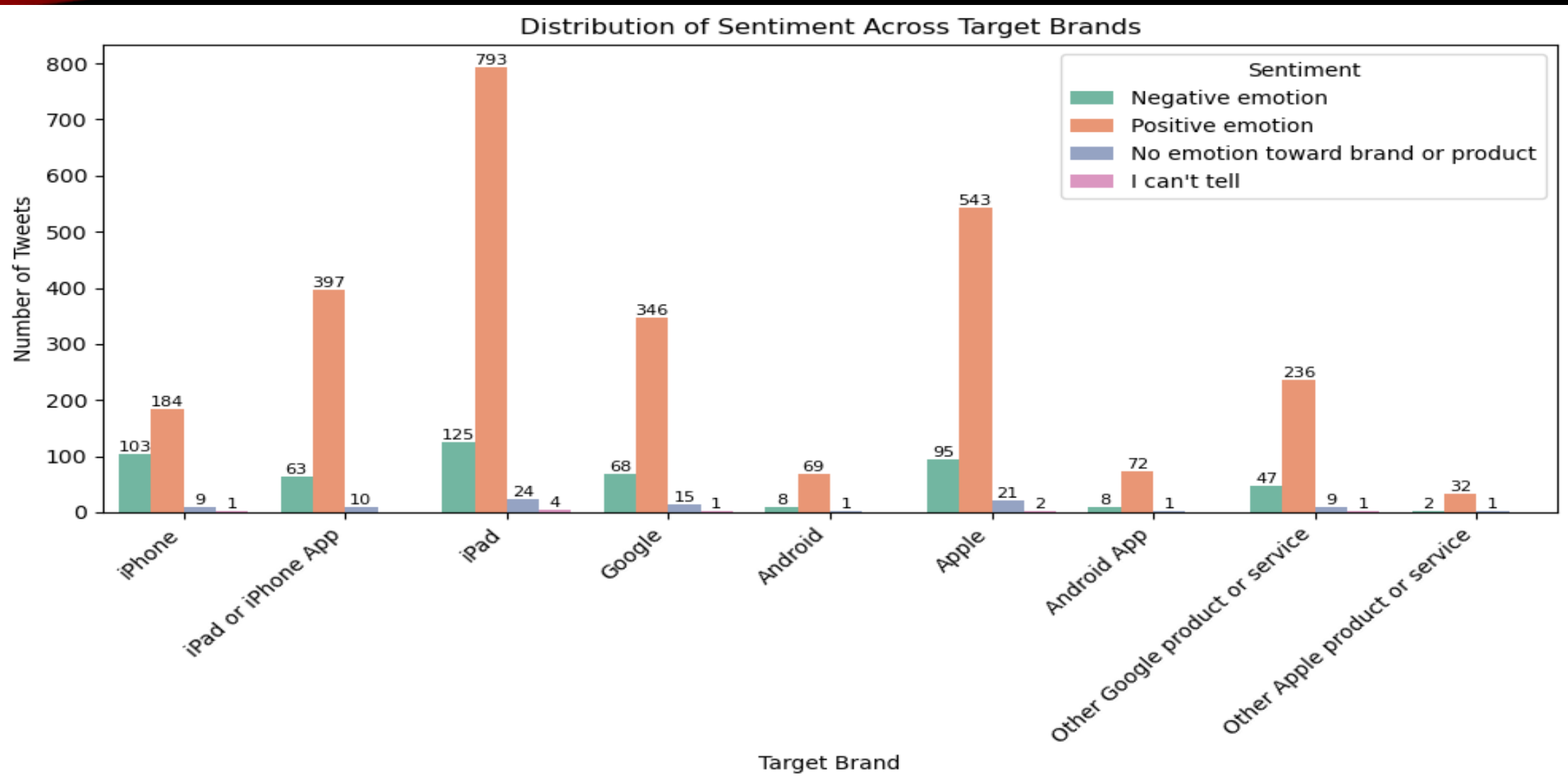
- ✓ Reduce noise
- ✓ Improve feature quality
- ✓ Better model performance

EXPLORATORY DATA ANALYSIS

Distribution of Tweet Sentiments

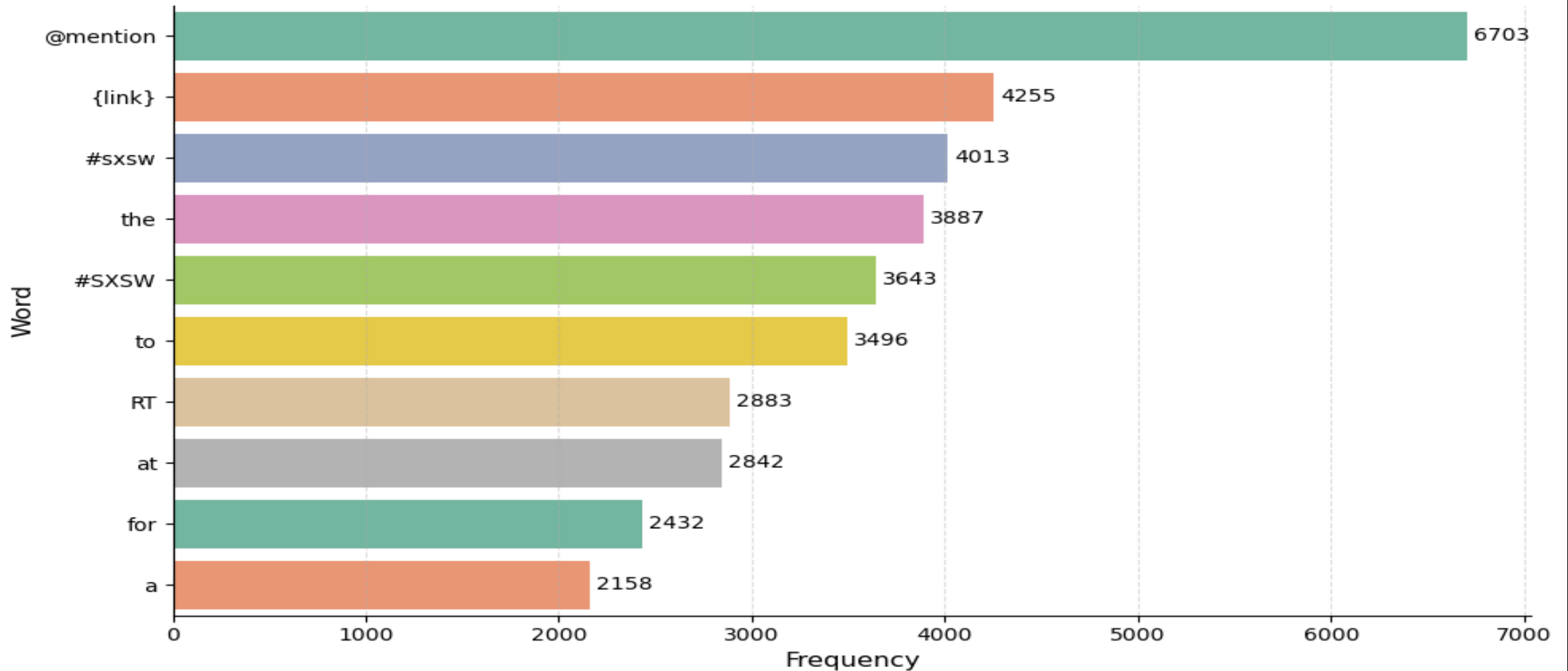


BRAND ANALYSIS



FREQUENT WORDS

Top 10 Most Frequent Words



FEATURE ENGINEERING

Transforming Text into Numbers

- TF-IDF Vectorization

Converts words into numerical features.

One-Hot Encoding

- Brand Feature:
- Apple $\rightarrow [1,0]$
- Google $\rightarrow [0,1]$

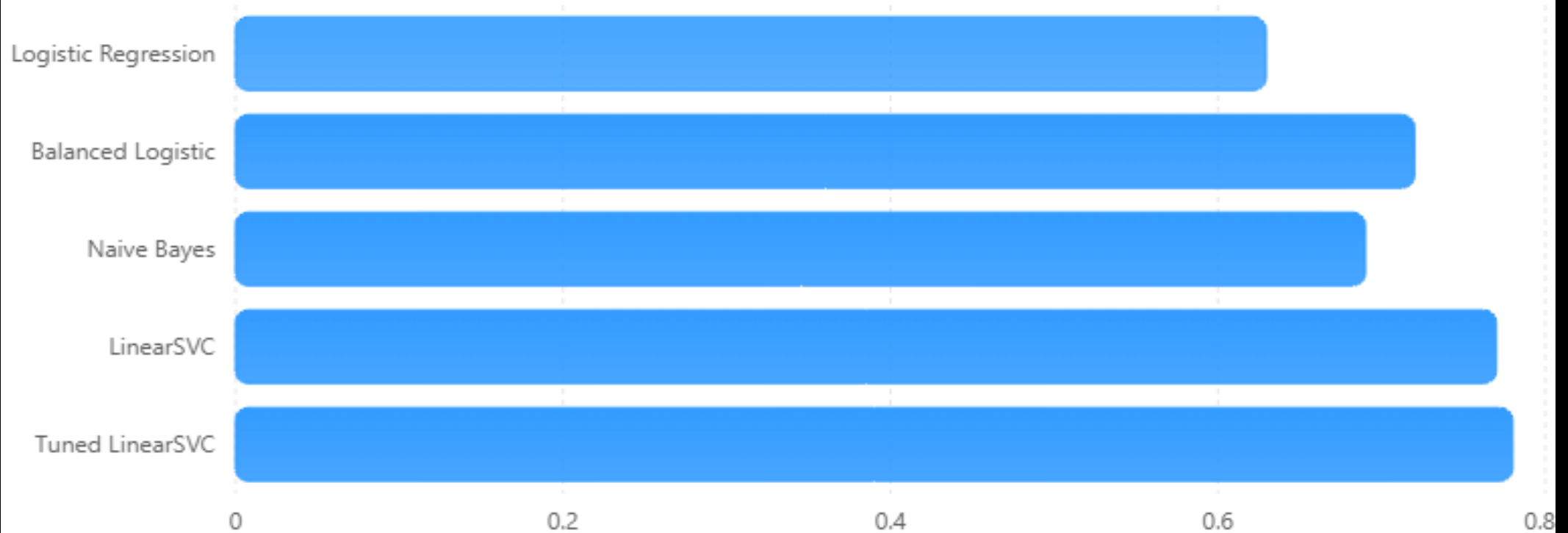
MACHINE LEARNING MODELS

- Models Evaluated
 -  Logistic Regression
 -  Balanced Logistic Regression
 -  Multinomial Naive Bayes
 -  LinearSVC
 -  Tuned LinearSVC
- Evaluation Metrics
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Macro F1

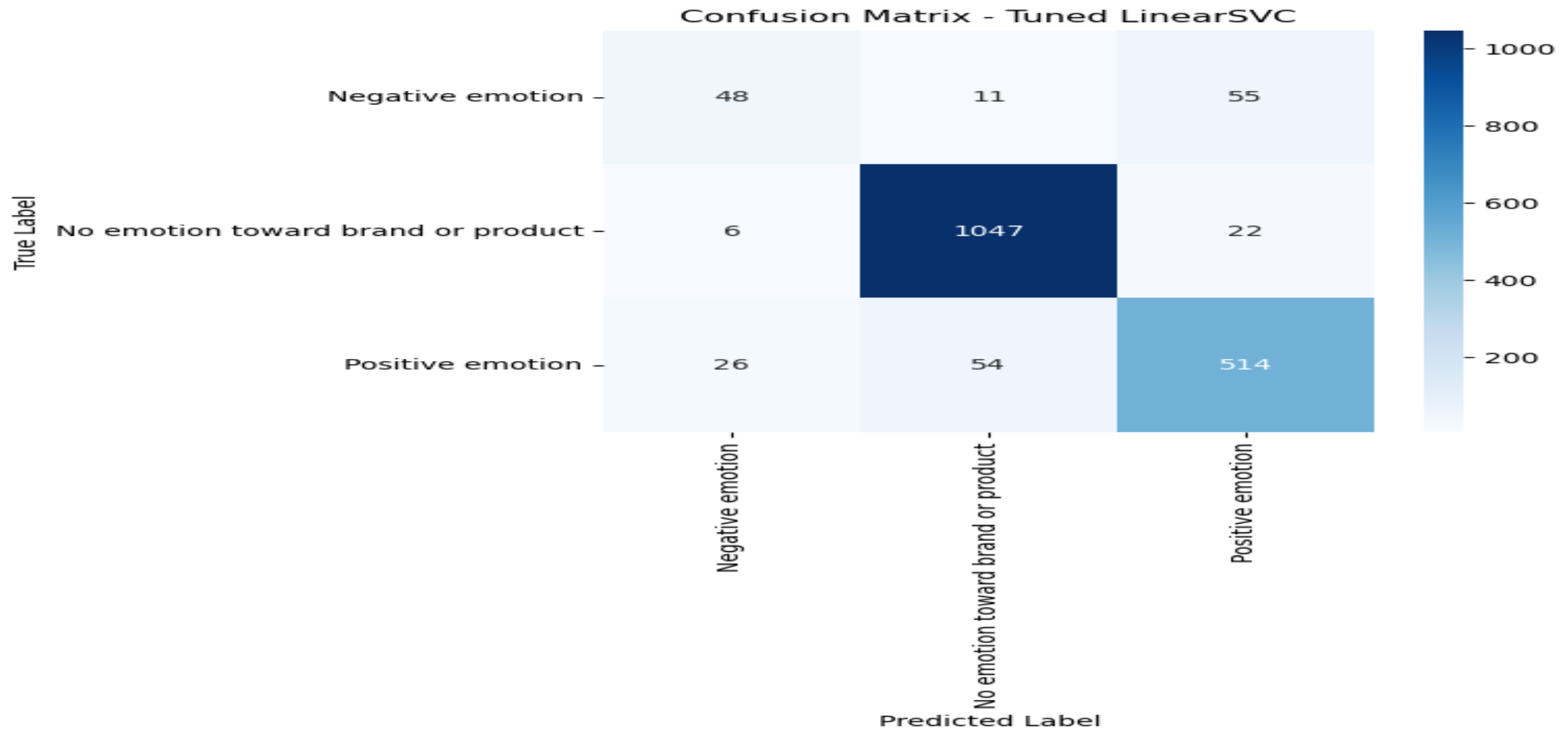
MODEL COMPARISON

Model Performance Comparison

Macro F1-score comparison across models.



BEST MODEL EVALUATION



BEST MODEL RESULTS

- Tuned LinearSVC
- Accuracy: 90.6%
- Macro F1: 0.78
- Why It Won
 - ✓ Strong neutral classification
 - ✓ Strong positive classification
 - ✓ Improved negative sentiment detection

MODEL EXPLAINABILITY

Understanding Predictions

- Feature Importance
- LIME Explanations

Insight

- The model's predictions can be explained using influential words and local explanations, improving transparency and trust.

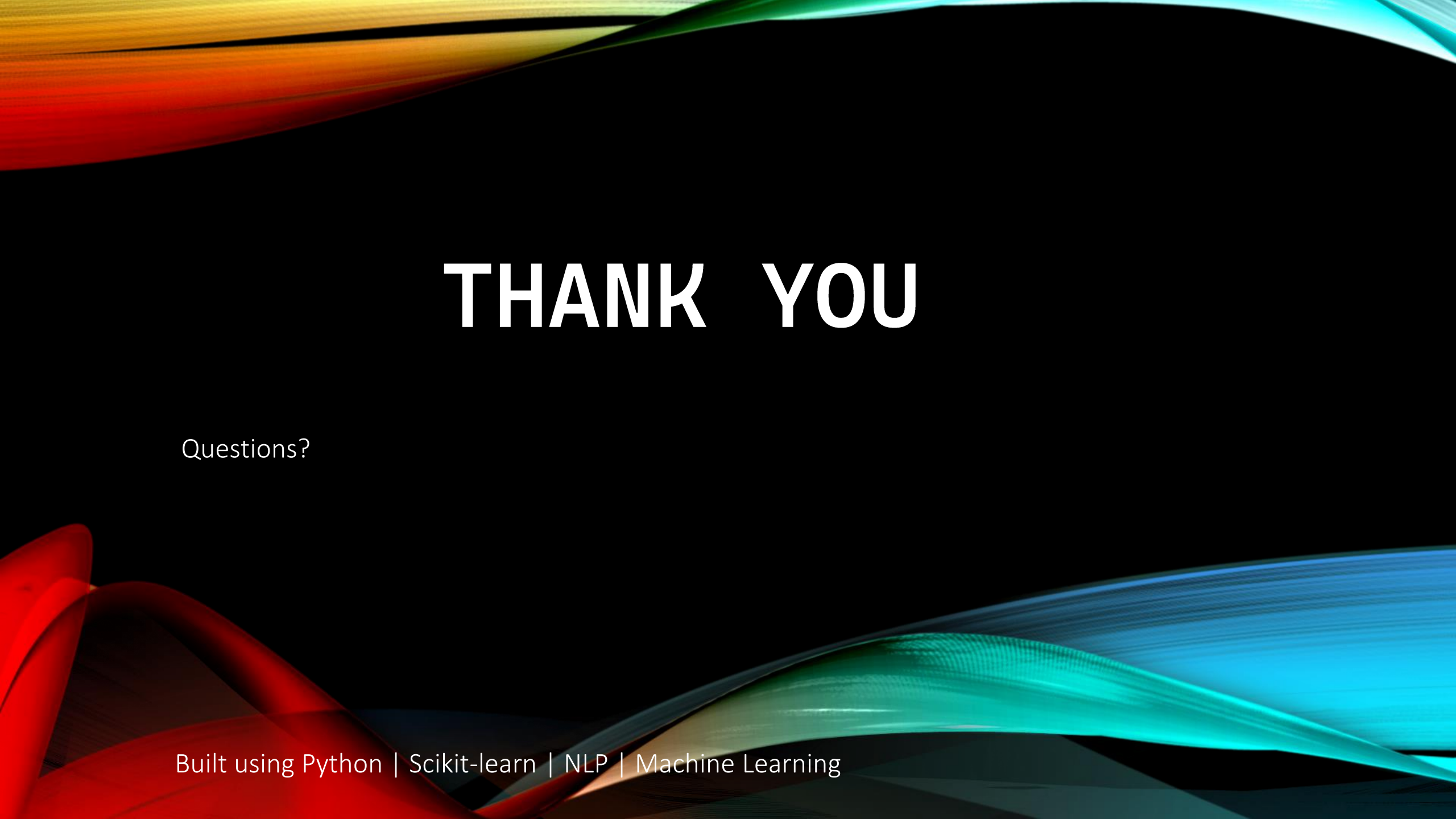
FINDINGS & RECOMMENDATIONS

Key Findings

-  Most tweets were neutral
-  Negative sentiment was underrepresented
-  Class imbalance affected baseline models
-  Tuned LinearSVC delivered the best performance

Recommendations

-  Deploy Tuned LinearSVC
-  Collect more negative sentiment examples
-  Retrain periodically
-  Explore BERT/Transformer models



THANK YOU

Questions?

Built using Python | Scikit-learn | NLP | Machine Learning